

Due: **Monday, Sep 28 by 5pm**

Reading: Ashcroft & Mermin, [Chaps 8, Chap 9](#); Kittel [Chap 2](#) (posted online); 8.511 classnotes, [part 1](#), [part 2](#); Steve Simon lecture notes, [Chap. 12](#) (active links)

Bloch bands and Fermi surfaces in space dimension $D > 1$.

Consider noninteracting spin- $\frac{1}{2}$ electrons moving in the two-dimensional periodic potential

$$U(x, y) = V_1 [\cos(Gx) + \cos(Gy)] + V_3 [\cos G(x + y) + \cos G(x - y)], \quad G = \frac{2\pi}{a}. \quad (1)$$

Assume throughout that the potential is weak,

$$|V_1|, |V_3| \ll \frac{\hbar^2 G^2}{2m}. \quad (2)$$

To leading order, the potential may therefore be neglected except near degeneracies of the empty-lattice bands. Spin does not affect the band dispersion in this problem, but it gives a factor of two in counting states and in the density of states.

The potential is unchanged under symmetry operations $x \rightarrow -x$, $y \rightarrow -y$, $x \leftrightarrow y$, and rotations by 90° . These elementary spatial symmetries will be useful below. A systematic treatment of crystal symmetries and their consequences for band degeneracies will be developed later in the course.

1. Empty-lattice bands

- (a) Determine the Bravais lattice in real space, its reciprocal lattice, and the first Brillouin zone. Sketch the first Brillouin zone and label the high-symmetry points

$$\Gamma = (0, 0), \quad X = (G/2, 0), \quad M = (G/2, G/2). \quad (3)$$

- (b) In the empty-lattice approximation, states at a crystal momentum $\mathbf{k}$ are obtained by folding free-electron states into the first Brillouin zone. Their energies are

$$\varepsilon_{\mathbf{G}'}^{(0)}(\mathbf{k}) = \frac{\hbar^2}{2m} |\mathbf{k} - \mathbf{G}'|^2, \quad (4)$$

where $\mathbf{G}'$ is a reciprocal-lattice vector. Identify the two degenerate states at X and the four degenerate states at M that originate from the lowest free-electron energy shell.

- (c) A Fourier component $U_{\mathbf{Q}}$ of the periodic potential couples plane waves whose wavevectors differ by $\mathbf{Q}$. List the nonzero Fourier components of the potential in Eq. (1). Which component couples the two states at X ? Which components couple the four states at M ?

2. Gap opening, critical points, and a Lifshitz transition

Consider momenta close to $X = (G/2, 0)$, the midpoint of the Brillouin-zone boundary $k_x = G/2$:

$$\mathbf{k} = \left(\frac{G}{2} + q_x, q_y \right), \quad |q_x|, |q_y| \ll G. \quad (5)$$

The two nearly degenerate free-electron states have wavevectors $\mathbf{k}$ and $\mathbf{k} - (G, 0)$.

- (a) Show that, in the basis formed by these two states, the Hamiltonian to leading order in the periodic potential is

$$H_X(\mathbf{q}) = \begin{pmatrix} \frac{\hbar^2}{2m} [(G/2 + q_x)^2 + q_y^2] & V_1/2 \\ V_1/2 & \frac{\hbar^2}{2m} [(-G/2 + q_x)^2 + q_y^2] \end{pmatrix}. \quad (6)$$

Explain why V_3 does not appear in this two-state Hamiltonian.

- (b) Find the two eigenvalues $E_+(\mathbf{q})$ and $E_-(\mathbf{q})$. Show that the periodic potential opens a gap at X , and determine the size of this gap.
- (c) Expand both eigenvalues through quadratic order in q_x and q_y . Write each expansion in the form

$$E(\mathbf{q}) - E(\mathbf{0}) = \frac{\hbar^2 q_x^2}{2m_x^*} + \frac{\hbar^2 q_y^2}{2m_y^*}. \quad (7)$$

Determine the signs of the two effective masses for each band in the weak-potential limit. Classify X as a local minimum, local maximum, or saddle point of each band.

- (d) The group velocity in band n is

$$\mathbf{v}_n(\mathbf{k}) = \frac{1}{\hbar} \nabla_{\mathbf{k}} E_n(\mathbf{k}). \quad (8)$$

Show from your answer above that the component of velocity normal to the zone boundary vanishes at X . Also explain this result using the fact that reflection symmetry and reciprocal-lattice periodicity imply

$$E_n(G/2 + q_x, q_y) = E_n(G/2 - q_x, q_y). \quad (9)$$

- (e) Sketch the constant-energy contours of the band that has a saddle point at X , for energies immediately below, at, and immediately above the saddle-point energy.

A *Lifshitz transition* is a change in the topology of a Fermi surface as the chemical potential or another control parameter passes through a critical energy of a band. It need not involve any spontaneous symmetry breaking. Describe the Lifshitz transition that occurs as the chemical potential passes through the saddle-point energy found above. In particular, explain how the Fermi contours reconnect.

3. Van Hove singularities in two dimensions

A critical point of a band is a momentum at which $\nabla_{\mathbf{k}} E(\mathbf{k}) = 0$. Near a nondegenerate critical point, suppose that the dispersion can be approximated by

$$E(\mathbf{q}) - E_c = \frac{\hbar^2 q_x^2}{2m_x} + \frac{\hbar^2 q_y^2}{2m_y}. \quad (10)$$

As illustrated in Problem 2, m_x and m_y may have either the same or opposite signs. The density of states per unit area, including spin degeneracy, is

$$\nu(E) = 2 \int \frac{d^2 q}{(2\pi)^2} \delta(E - E(\mathbf{q})). \quad (11)$$

- (a) Calculate the nonanalytic contribution to $\nu(E)$ when $m_x m_y > 0$, corresponding to a local minimum or maximum. Sketch the result on both sides of E_c .
- (b) Calculate the singular contribution to $\nu(E)$ when $m_x m_y < 0$, corresponding to a saddle point. The quadratic approximation must be restricted by a momentum or energy cutoff; the precise value of this cutoff is not important. Show that the singularity is logarithmic and determine its prefactor.
- (c) Give a geometric explanation, in terms of the constant-energy contours, for why a saddle point produces a stronger singularity than an ordinary minimum or maximum in two dimensions. Relate your answer to the Lifshitz transition in Problem 2.

4. Optional challenge: a four-state degeneracy at M

At $M = (G/2, G/2)$, consider the four degenerate empty-lattice states labeled by the reciprocal-lattice vectors

$$(0, 0), \quad (G, 0), \quad (0, G), \quad (G, G). \quad (12)$$

- (a) Verify directly that these four states have the same free-electron energy at M .
- (b) Using the Fourier components found in Problem 1, show that the perturbation within this four-dimensional subspace is

$$H_M - E_M^{(0)} I = \frac{1}{2} \begin{pmatrix} 0 & V_1 & V_1 & V_3 \\ V_1 & 0 & V_3 & V_1 \\ V_1 & V_3 & 0 & V_1 \\ V_3 & V_1 & V_1 & 0 \end{pmatrix}. \quad (13)$$

- (c) Find the four energy shifts and their degeneracies. One pair of states remains degenerate for arbitrary values of V_1 and V_3 . Explain intuitively why the square symmetry of the potential makes such a degeneracy plausible. A systematic explanation of this result will be given later using the theory of crystal symmetries.